

二次域练习题

设 $D \in \mathbb{Z}$ 是无平方因子的非零整数, $D \neq 1$

(iv) 当 $D \equiv 1 \pmod{4}$ 时, $O_D := \left\{ \frac{x}{2} + \frac{y}{2}\sqrt{D} \mid x, y \in \mathbb{Z} \text{ 且 } x \equiv y \pmod{2} \right\}$ 是整环

Proof: $\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \in \mathbb{C} : a, b \in \mathbb{Q}\}$ 是整环, $O_D \subseteq \mathbb{Q}(\sqrt{D})$,

$$\because 0 = \frac{0}{2} + \frac{0}{2}\sqrt{D} \in O_D \quad \therefore O_D \neq \emptyset$$

$$0 = \frac{0}{2} + \frac{0}{2}\sqrt{D} \in O_D, \quad 1 = \frac{2}{2} + \frac{0}{2}\sqrt{D} \in O_D$$

对 $\forall \frac{x_1}{2} + \frac{y_1}{2}\sqrt{D}, \frac{x_2}{2} + \frac{y_2}{2}\sqrt{D} \in O_D$ (其中 $x_1, y_1 \in \mathbb{Z}, x_1 \equiv y_1 \pmod{2}, x_2, y_2 \in \mathbb{Z}, x_2 \equiv y_2 \pmod{2}$), 有:

$$\left(\frac{x_1}{2} + \frac{y_1}{2}\sqrt{D}\right) + \left(\frac{x_2}{2} + \frac{y_2}{2}\sqrt{D}\right) = \frac{x_1+x_2}{2} + \frac{y_1+y_2}{2}\sqrt{D} \in O_D$$

$$(\because x_1, x_2 \in \mathbb{Z} \quad \therefore x_1+x_2 \in \mathbb{Z} \quad \because y_1, y_2 \in \mathbb{Z} \quad \therefore y_1+y_2 \in \mathbb{Z})$$

$$\because x_1 \equiv y_1 \pmod{2}, \quad x_2 \equiv y_2 \pmod{2} \quad \therefore x_1+x_2 \equiv y_1+y_2 \pmod{2} \quad)$$

$$\left(\frac{x_1}{2} + \frac{y_1}{2}\sqrt{D}\right) \cdot \left(\frac{x_2}{2} + \frac{y_2}{2}\sqrt{D}\right) = \frac{\frac{x_1x_2+y_1y_2D}{2}}{2} + \frac{\frac{x_1y_2+x_2y_1}{2}}{2}\sqrt{D} \in O_D$$

$$\because x_1 \equiv y_1 \pmod{2} \quad \therefore 2 \mid x_1 - y_1 \quad \therefore x_1 - y_1 = 2k_1 \quad (k_1 \in \mathbb{Z}) \quad \therefore x_1 = y_1 + 2k_1$$

$$\because x_2 \equiv y_2 \pmod{2} \quad \therefore 2 \mid x_2 - y_2 \quad \therefore x_2 - y_2 = 2k_2 \quad (k_2 \in \mathbb{Z}) \quad \therefore x_2 = y_2 + 2k_2$$

$$\because D \equiv 1 \pmod{4} \quad \therefore 4 \mid D-1 \quad \therefore D-1 = 4\lambda \quad (\lambda \in \mathbb{Z}) \quad \therefore D = 4\lambda + 1$$

$$\therefore \frac{x_1x_2+y_1y_2D}{2} = y_1y_2(2\lambda+1) + k_2y_1 + k_1y_2 + 2k_1k_2 \in \mathbb{Z}$$

$$\frac{x_1y_2+x_2y_1}{2} = y_1y_2 + k_1y_2 + k_2y_1 \in \mathbb{Z}$$

$$\frac{x_1x_2+y_1y_2D}{2} - \frac{x_1y_2+x_2y_1}{2} = 2(\lambda y_1y_2 + k_1k_2)$$

$$\therefore \frac{x_1x_2+y_1y_2D}{2} \equiv \frac{x_1y_2+x_2y_1}{2} \pmod{2}$$

对 $\forall \frac{x}{2} + \frac{y}{2}\sqrt{D} \in O_D$ (其中 $x, y \in \mathbb{Z}$ 且 $x \equiv y \pmod{2}$)

$$\therefore \left(\frac{x}{2} + \frac{y}{2}\sqrt{D}\right) + \left(\frac{-x}{2} + \frac{-y}{2}\sqrt{D}\right) = \frac{0}{2} + \frac{0}{2}\sqrt{D} = 0$$

$$\therefore -\left(\frac{x}{2} + \frac{y}{2}\sqrt{D}\right) = \frac{-x}{2} + \frac{-y}{2}\sqrt{D} \in O_D$$

$$\left(\begin{array}{l} \because x \in \mathbb{Z} \quad \therefore -x \in \mathbb{Z} \quad \because y \in \mathbb{Z} \quad \therefore -y \in \mathbb{Z} \\ \therefore x \equiv y \pmod{2} \quad \therefore -x \equiv -y \pmod{2} \end{array} \right)$$

$\therefore O_D$ 是 $\mathbb{Q}(\sqrt{D})$ 的子环 $\therefore O_D$ 是整环 □

(v) 当 $D \equiv 1 \pmod{4}$ 时, $\mathbb{Z}[\sqrt{D}] \subseteq O_D$

Proof: 对 $\forall a+b\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$ (其中 $a, b \in \mathbb{Z}$), 有:

$$a+b\sqrt{D} = \frac{2a}{2} + \frac{2b}{2}\sqrt{D} \in O_D$$

$$\therefore a \in \mathbb{Z} \quad \therefore 2a \in \mathbb{Z} \quad \because b \in \mathbb{Z} \quad \therefore 2b \in \mathbb{Z}$$

$$\therefore 2a - 2b = 2(a-b) \quad \therefore \frac{2a-2b}{2} = a-b \in \mathbb{Z} \quad \therefore 2 \mid 2a-2b$$

$$\therefore 2a \equiv 2b \pmod{2}$$

$$\therefore \mathbb{Z}[\sqrt{D}] \subseteq O_D.$$

(vi) 对 $\forall D \in \mathbb{Z}$ 是无平方因子的非零整数, $D \neq 1$, 有: $\mathbb{Z}[\sqrt{D}] \subseteq O_D$

Proof: 对 $\forall a+b\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$ (其中 $a, b \in \mathbb{Z}$), 有:

$$a+b\sqrt{D} = \frac{2a}{2} + \frac{2b}{2}\sqrt{D} \in O_D$$

$$2a \in \mathbb{Z}, \quad 2b \in \mathbb{Z}, \quad 2a-2b = 2(a-b), \quad 2 \mid 2a-2b \quad \therefore 2a \equiv 2b \pmod{2}$$

$$\therefore \mathbb{Z}[\sqrt{D}] \subseteq O_D. \quad \square$$